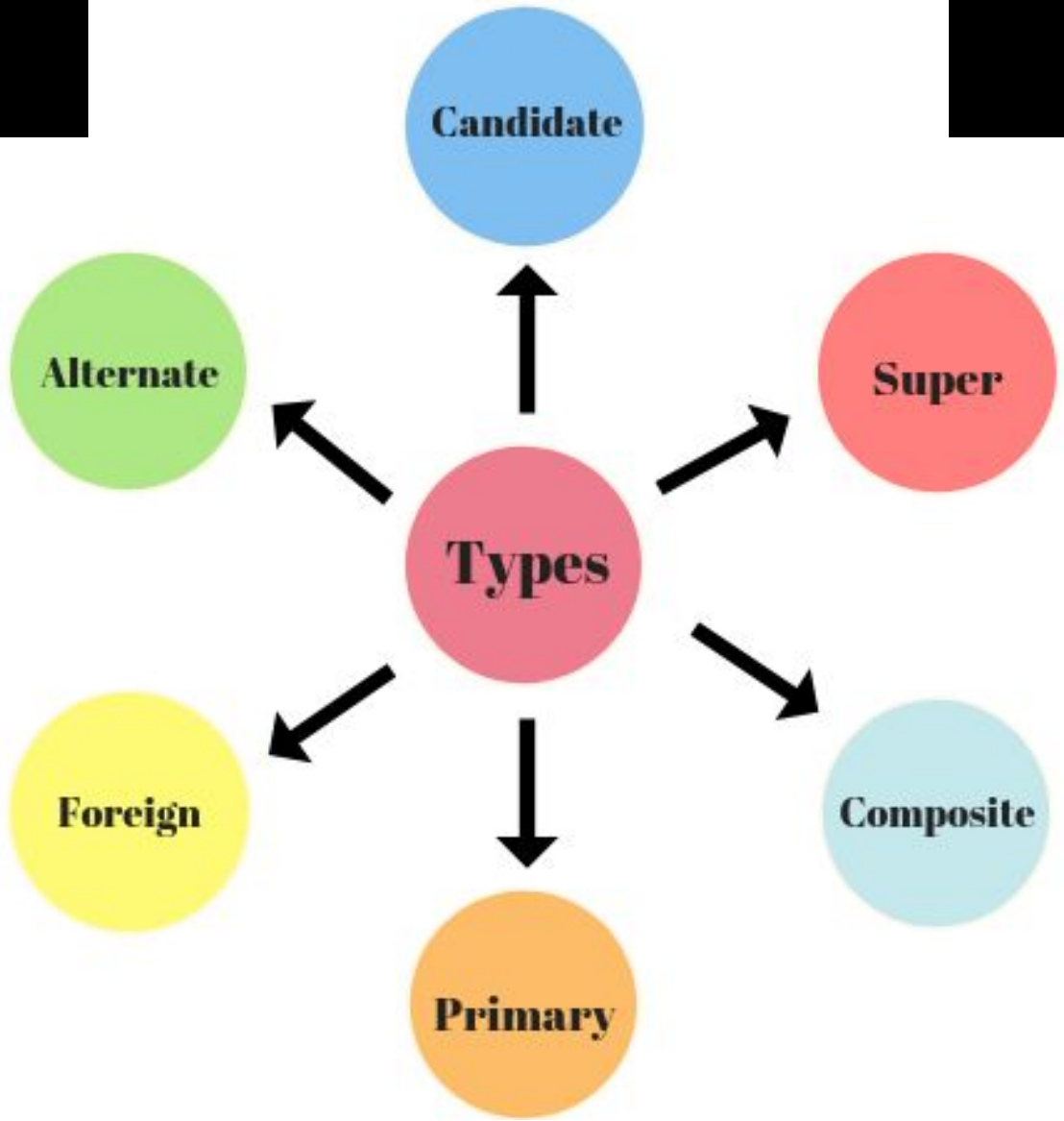
Keys in DBMS

- Keys play an important role in the relational database.
  - It is used to uniquely identify any record or row of data from the table.
- It is used to establish and identify relationships between tables.

Types of keys:

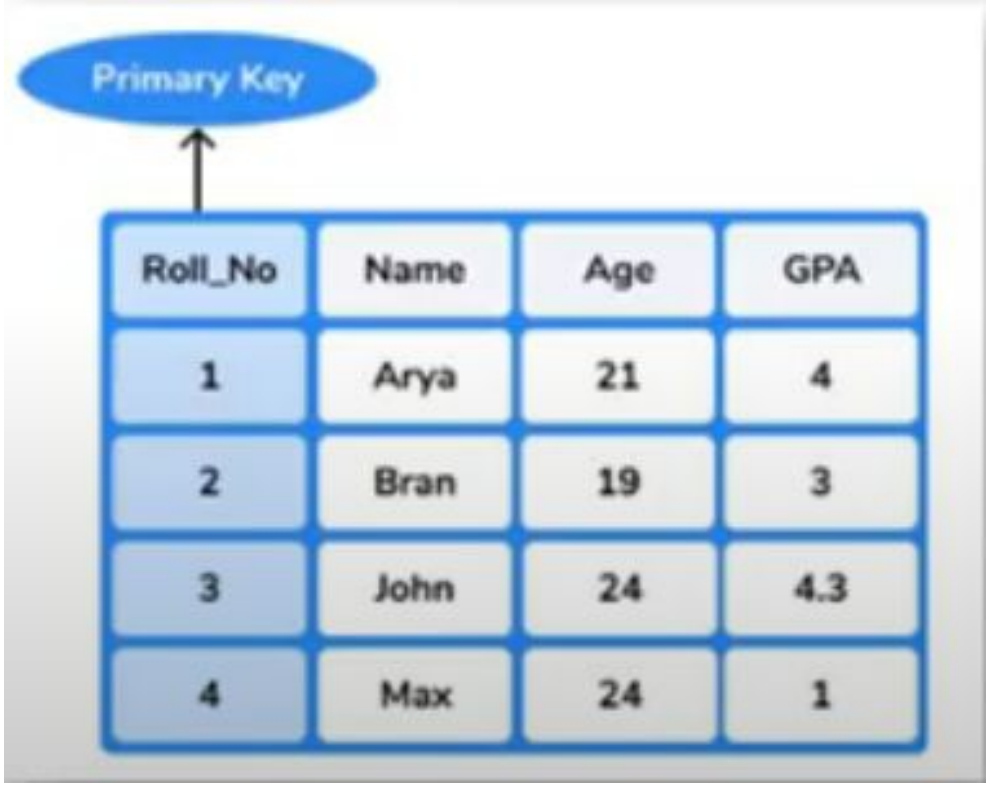


1 Primary key

- It is the first key used to identify one and only one instance of an entity uniquely.

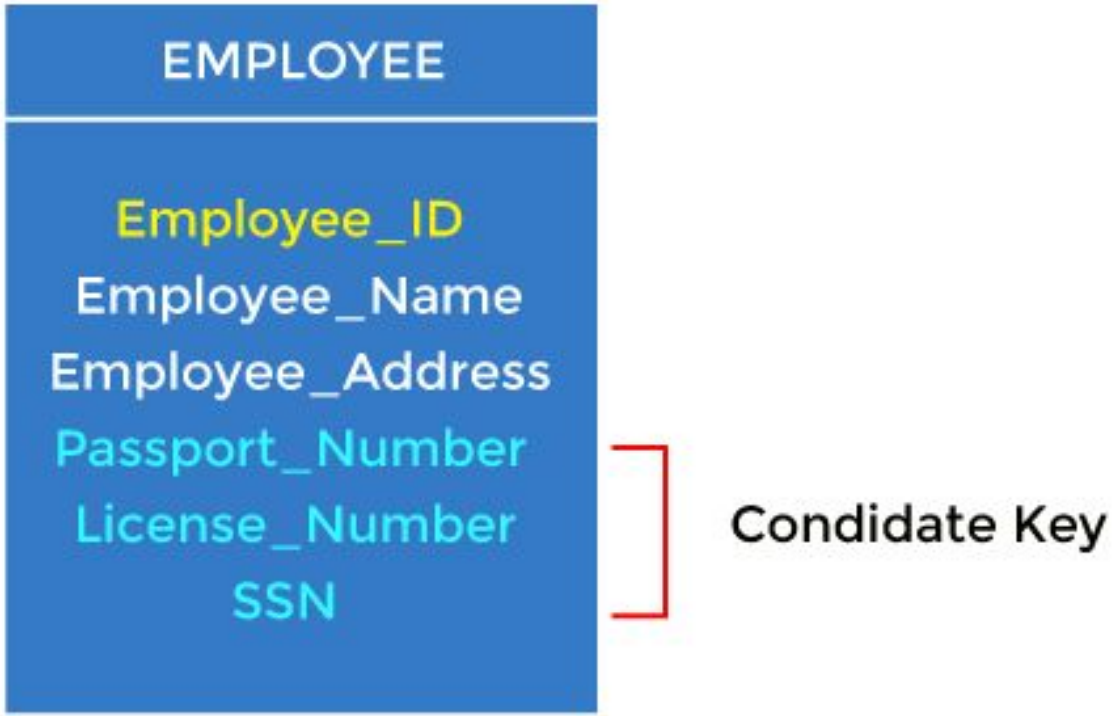
Rules for defining Primary Key:

1. A **primary key cannot be <u>NULL</u>**
2. It must have **<u>unique values</u>** <u>and cannot contain any **duplicate** values.</u>
3. **The value of Primary key <u>can never be changed</u>.**
4. **The value of Primary key must be <u>assigned when inserting a record.</u>**
5. **<u>Two rows can't have the same primary key value.</u>**


Primary Key
Roll_No
Name
Age
GPA
1
Arya
21
4
2
Bran
19
3
3
John
24
4.3
4
Max
24
1

2. **Candidate Key**

- A candidate key is an attribute or set of attributes that can uniquely identify a tuple.
- A table can have multiple candidate keys but only one primary key.
- A primary key should be selected from the candidate keys.
- The candidate keys are as strong as the primary key.

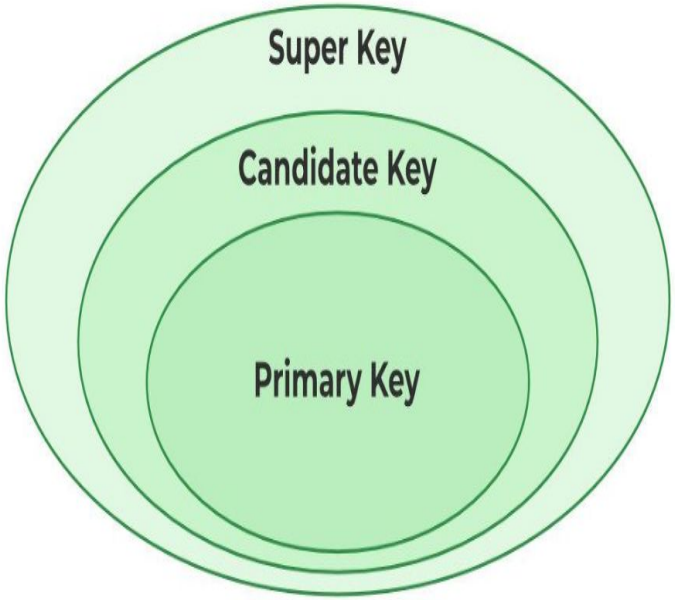

EMPLOYEE
Employee_ID
Employee_Name
Employee_Address
Passport_Number
License_Number
SSN
Condidate Key

3. Super Key

- Super key is an attribute set that can uniquely identify a tuple.
- A super key is not restricted to have any specific number of attributes.
- A super key is a group of single or multiple keys that uniquely identifies rows in a table.
- It supports NULL values in rows.

3. **Super Key**

- Rules for defining Super key:
- Adding zero or more attribute to the candidate
- A candidate key is a super key but vice versa is not true.
- It supports Null Value



➢**Example:**

Student (roll , name , sex , age , address , class , section)

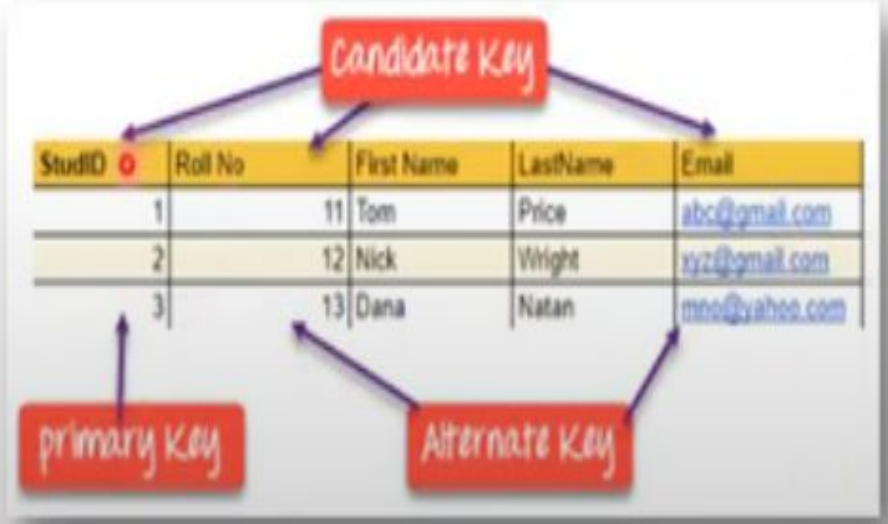
- (roll , name , sex , age , address , class , section)
- (class , section , roll)
- (class , section , roll , sex)
- (name , address)

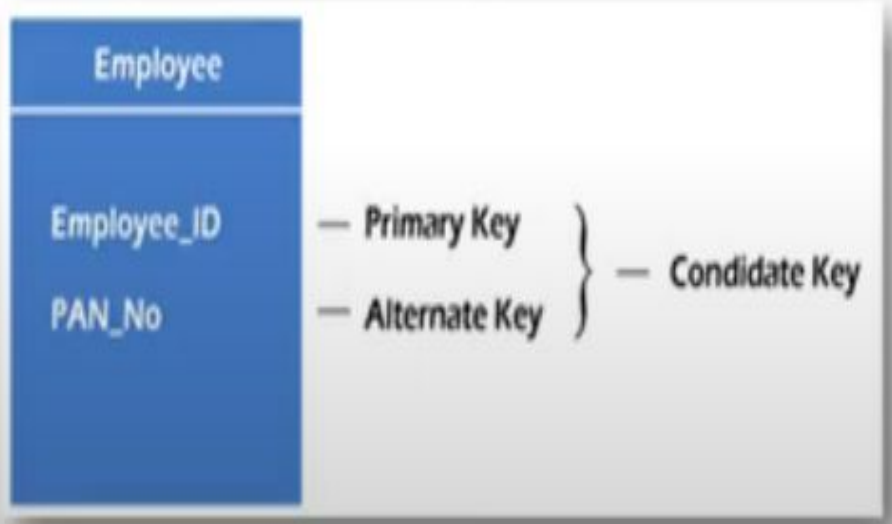
- Each set can uniquely identify each student in the Student table.

4. Alternate Key

- Alternate Keys is uniquely identify every row in that table.
- All the keys which are not primary key are called an Alternate Key.
- If there is only one candidate key in a relation, it does not have an alternate key.

➢**Total number of the Alternate Keys = Total number of Candidate Keys - Primary Key.**





5. Foreign Key

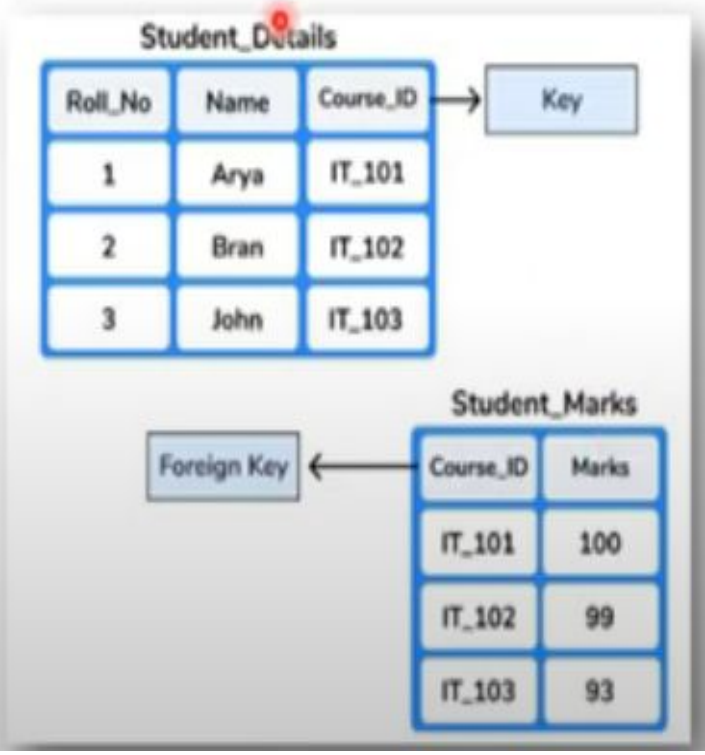
- Foreign keys is a column that creates a relationship between two tables.
- Foreign keys are the column of the table used to point to the primary key of another table.
- It is a key it acts as a primary key in one table & secondary key in another table.
- The purpose of Foreign keys is to maintain data integrity and allow navigation between two different instances of an entity

➢**Rules for defining Foreign Key:**

1. Foreign key references the primary key of the table.
2. Foreign key can take the NULL value.
3. There is no restriction on a foreign key to be unique.

✓**Student Details:** Referencing relation/ Foreign Table.

✓**Student Marks:** Referenced relation / Primary or Master Table.



Composite key

- A **composite key** in DBMS is a **combination of two or more columns** that uniquely identify a row in a table. A composite key is used when a single column is not sufficient to uniquely distinguish records.

Example of Composite Key

Consider a **Student_Courses** table that records which students enroll in which courses:

Student_ID	Course_ID	Enrollment_Date
101	CS101	2024-01-15
102	CS101	2024-01-16
101	CS102	2024-01-17

- Here, **Student_ID** alone cannot be a primary key because a student can enroll in multiple courses.

- **Course_ID** alone also cannot be a primary key because multiple students can enroll in the same course.

- But the **combination of (Student_ID, Course_ID)** uniquely identifies each row.

Thus, **(Student_ID, Course_ID) together form a composite key**.